cBioPortal
FOR CANCER GENOMICS

Tutorial 5
Onco Query Language (OQL)

Use OQL to refine your queries

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cbioportal.org/tutorials

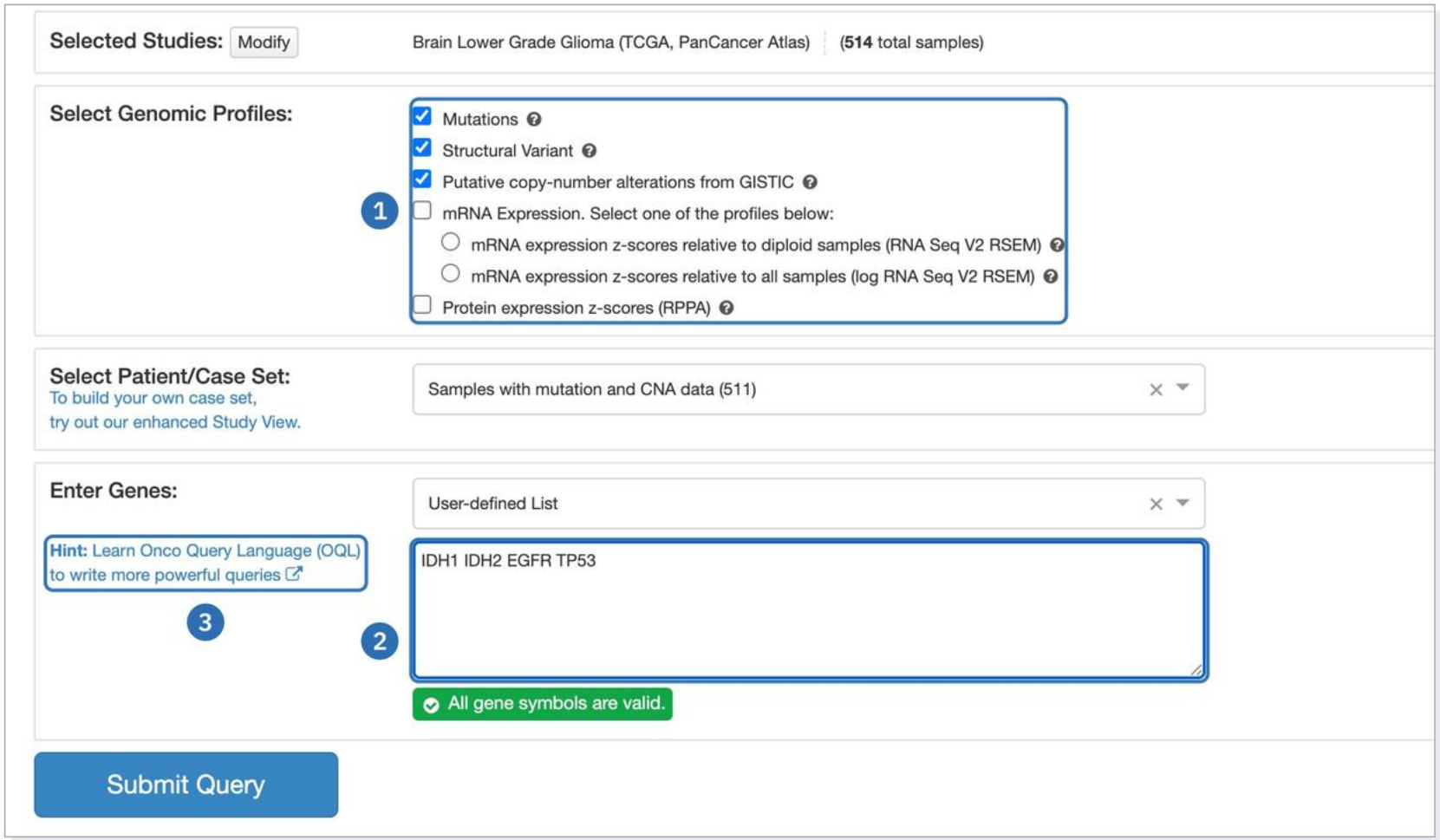
Tutorial objectives

- Introduce Onco Query Language (OQL)
- Explain the basic rules of OQL: keywords, modifiers, DATATYPES and gene tracks
- Show what OQL can do with three examples: glioma, BRCA1/2 inactivation in ovarian cancer, and RTK pathway alterations in colorectal cancer

What is OQL?

- OQL defines which kinds of alterations count when you query a gene: a sample is “altered” only if it has one of them.
- Without OQL, cBioPortal uses a default set. With OQL you can add alterations (e.g. gains, low expression) or narrow the query (e.g. one mutation, only drivers, only germline).
- Example: IDH1, IDH2, EGFR and TP53 in Brain Lower Grade Glioma (TCGA, PanCancer Atlas).

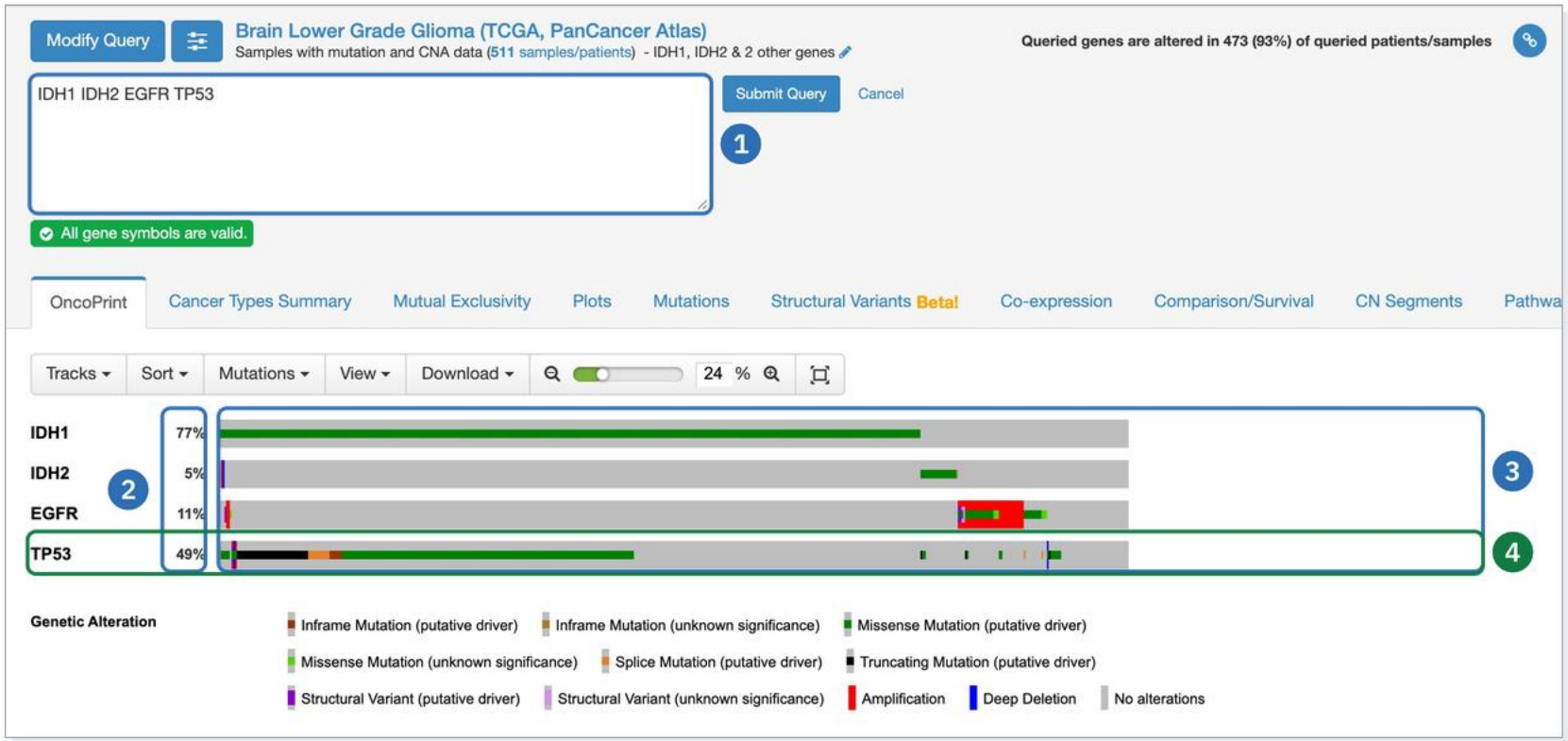
A regular query



1. Mutations, structural variants and copy number are the profiles queried by default.
2. Enter the genes: IDH1 IDH2 EGFR TP53 (no OQL yet).
3. The “Hint: Learn Onco Query Language (OQL) to write more powerful queries” link opens the OQL documentation.

Link to this page

A regular query: what counts as altered?



1. The query as typed: just the gene names.
2. 93% of samples are altered: IDH1 77%, IDH2 5%, EGFR 11%, TP53 49%.
3. A sample counts as altered if it has a mutation, a structural variant, an amplification or a deep deletion.
4. Q: Are there samples with shallow deletions in these genes?
   A: We can't tell from this query. Without OQL, only the default alterations are queried (MUT FUSION AMP HOMDEL), so shallow deletions (HETLOSS) are not shown, e.g. in TP53.

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The rules of OQL: keywords

	Data Type	Keywords and Syntax	Default*
1	Mutations	`MUT` All non-synonymous mutations `MUT = <protein change>` Specific amino acid changes (e.g. `V600E` or `V600`) `MUT = <mutation type>` Acceptable values are: `MISSENSE, NONSENSE, NONSTART, NONSTOP, FRAMESHIFT, INFRAME, SPLICE, TRUNC`	`MUT`
2	Fusions	`FUSION` All fusions (note that many studies lack fusion data)	`FUSION`
3	Copy Number Alterations	`AMP` Amplifications `HOMDEL` Deep Deletions `GAIN` Gains `HETLOSS` Shallow Deletions Comparison operators can also be used with `CNA` (e.g. `CNA >= GAIN` is the same as `AMP GAIN`)	`AMP` `HOMDEL`
4	mRNA Expression	`EXP < -x` mRNA expression is less than `x` standard deviations (SD) below the mean `EXP > x` mRNA expression is greater than `x` SD above the mean The comparison operators `<=` and `>=` also work	`EXP >= 2` `EXP <= -2`
4	Protein/phosphoprotein level	`PROT < -x` Protein expression is less than `x` standard deviations (SD) below the mean `PROT > x` Protein expression is greater than `x` SD above the mean The comparison operators `<=` and `>=` also work	`PROT >= 2` `PROT <= -2`

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1. Mutations: MUT, a specific change (MUT = V600E) or position (MUT = V600), or a type (MUT = MISSENSE, TRUNC...).
2. Structural variants: FUSION.
3. Copy number: AMP, HOMDEL (deep deletion), GAIN, HETLOSS (shallow deletion), or CNA >= GAIN.
4. mRNA and protein: EXP or PROT above or below a number of standard deviations, e.g. EXP < -2. The matching profile must be selected in the query form.
5. The defaults, used when a gene has no OQL: MUT FUSION AMP HOMDEL, and EXP/PROT >= 2 or <= -2.

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The rules of OQL: modifiers and syntax

- GENE: KEYWORD1 KEYWORD2 ... one gene per line (or separated by ;). The order of keywords does not matter.
- DRIVER: only putative drivers (as set in the settings menu: OncoKB, hotspots). Use it alone (EGFR: DRIVER) or joined to a keyword (MUT_DRIVER, AMP_DRIVER).
- GERMLINE and SOMATIC: only germline, or only somatic, mutations (MUT_GERMLINE). Only a few studies in the public portal have germline data.
- (a-b): mutations in protein positions a to b, e.g. MUT = (130-140).
- DATATYPES: KEYWORDS; sets the same OQL for all the genes after it.
- ["name" GENE1 GENE2 ...] merges genes into one track in the OncoPrint.

Full specification: docs.cbioportal.org/user-guide/oql

Using OQL: the same query, written out

- IDH1: MUT FUSION AMP HOMDEL (one line per gene: gene, colon, keywords)
- DATATYPES: MUT FUSION AMP HOMDEL; IDH1 IDH2 EGFR TP53 (the same keywords for all genes at once)
- Both give exactly the regular query: 93% of samples altered. The next slides change one thing at a time.

Using OQL: DATATYPES



1. DATATYPES: MUT FUSION AMP HOMDEL; then the genes. The same keywords apply to all of them.

2. The same result as the regular query: IDH1 77%, IDH2 5%, EGFR 11%, TP53 49%. Hover over a gene name to see its OQL.

Using OQL: add gains and shallow deletions



1. Add GAIN to EGFR and HETLOSS to TP53.
2. EGFR is now altered in 25% of samples (was 11%) and TP53 in 50%. Gains are light red, shallow deletions light blue.

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Using OQL: one mutation



1. IDH1: MUT=R132C, only this IDH1 mutation.
2. IDH1 is now altered in 3% of samples (was 77%): most IDH1 mutations in this study are R132H.

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Using OQL: leave out what doesn't matter



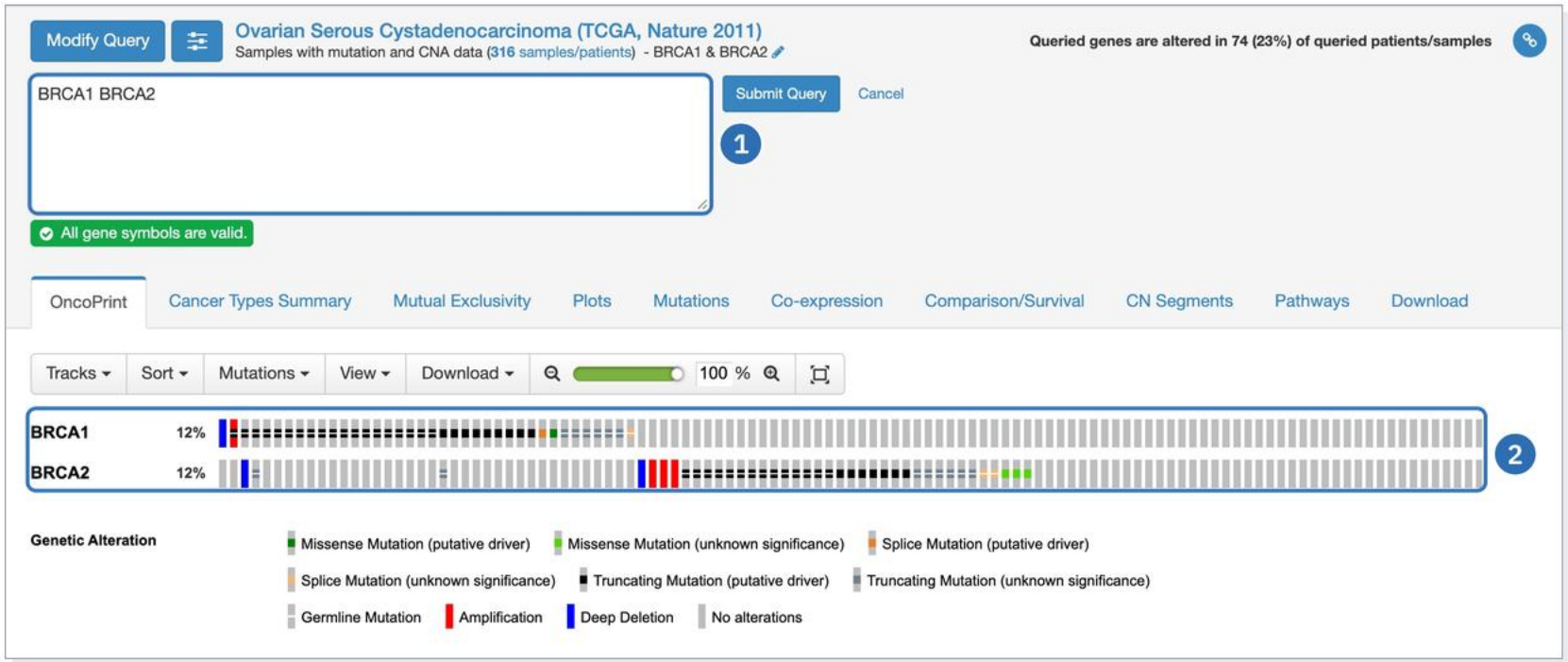
1. Remove alteration types that are not biologically relevant: here only mutations for IDH2, and no deep deletions for EGFR.

2. IDH2 drops to 4% (its structural variants are gone). EGFR is unchanged: it has no deep deletions.

OQL example: BRCA1/2 inactivation in ovarian cancer

- Loss of BRCA1 or BRCA2 is common in ovarian cancer. How many samples lose them?
- Query BRCA1 BRCA2 in Ovarian Serous Cystadenocarcinoma (TCGA, Nature 2011), samples with mutation and CNA data (316).

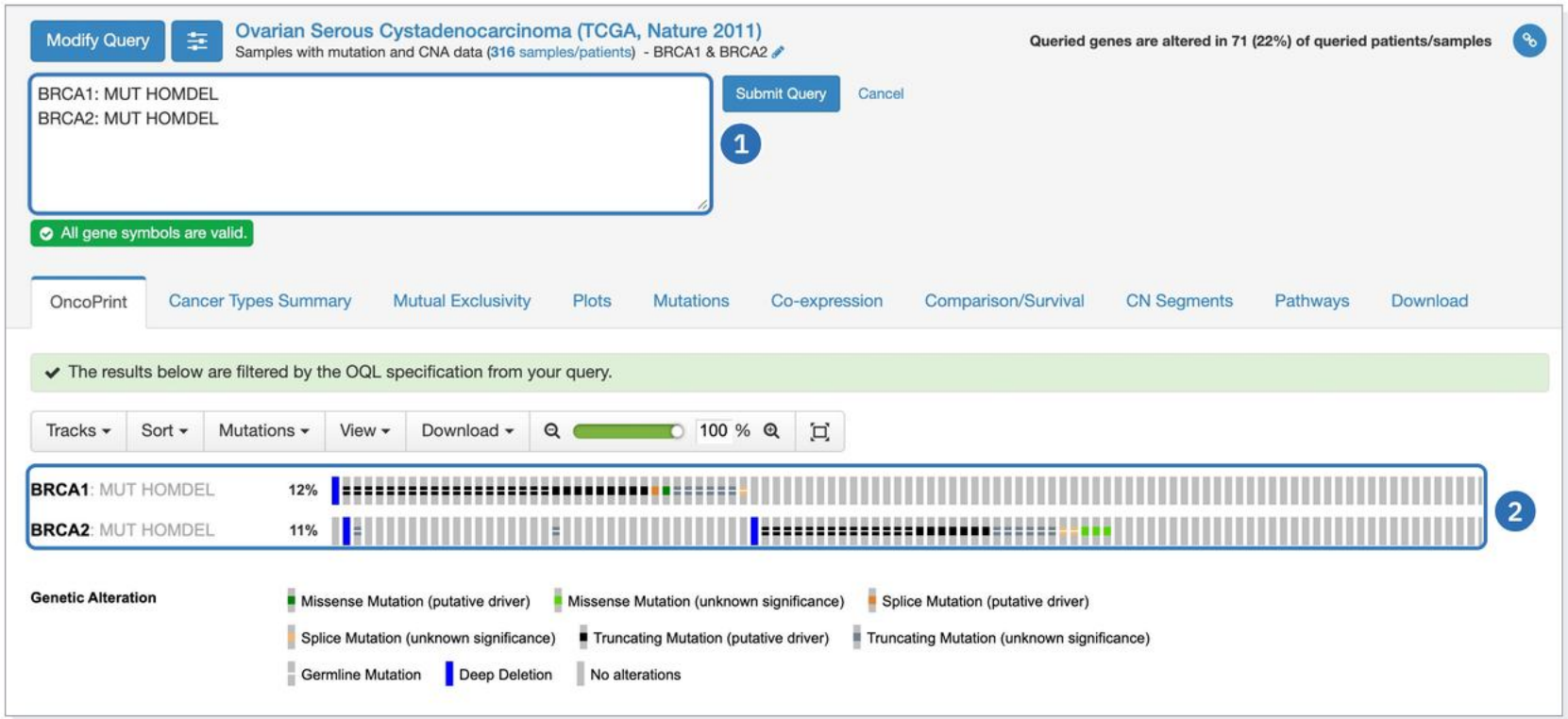
BRCA1/2: the regular query



1. BRCA1 BRCA2, no OQL.
2. 12% of samples are altered in each gene. But this includes amplifications (red), which do not cause a loss of function.

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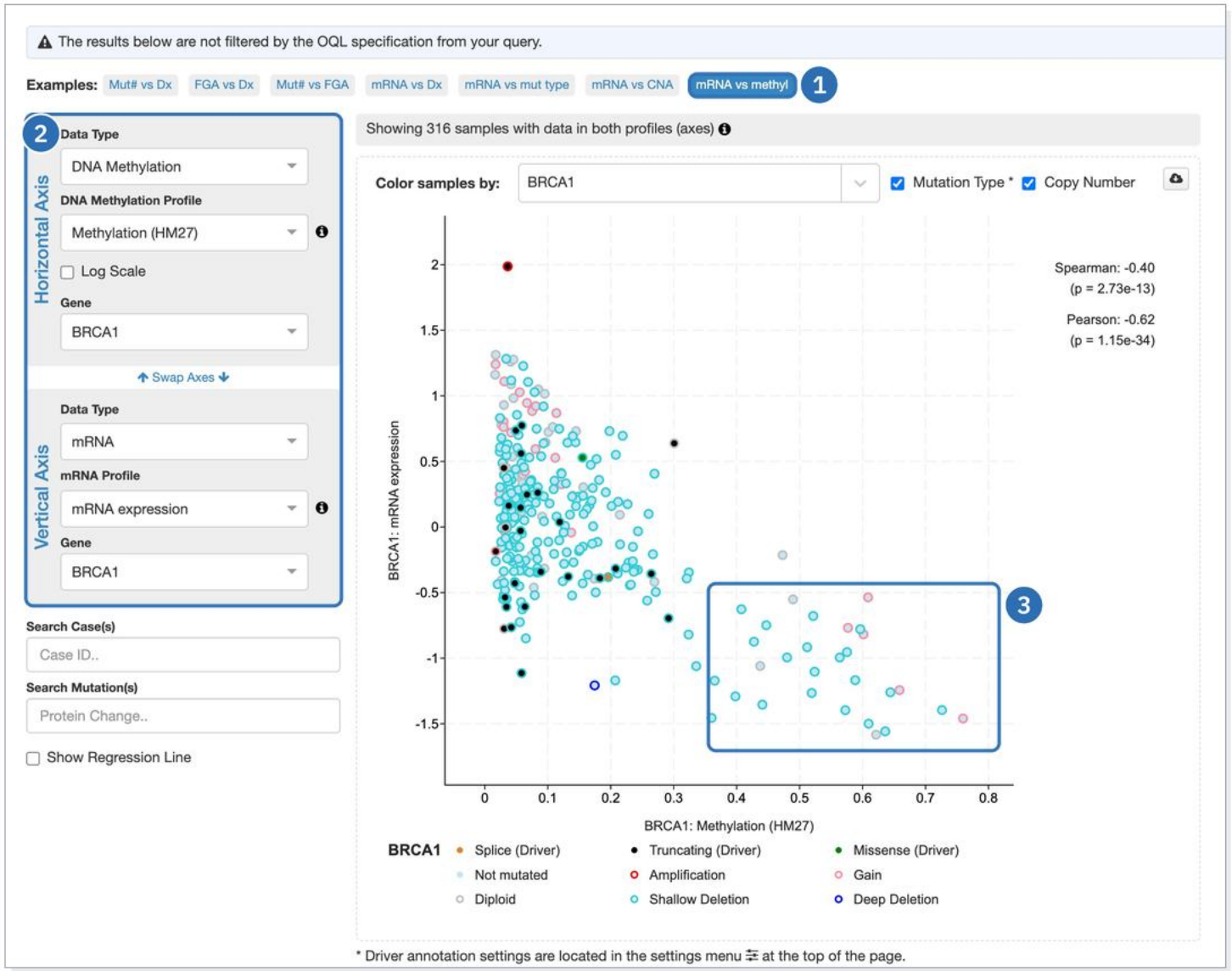
BRCA1/2: only mutations and deep deletions



1. BRCA1: MUT HOMDEL and BRCA2: MUT HOMDEL.
2. A better estimate of loss: BRCA1 12%, BRCA2 11% (22% of samples altered in either BRCA1 or BRCA2).

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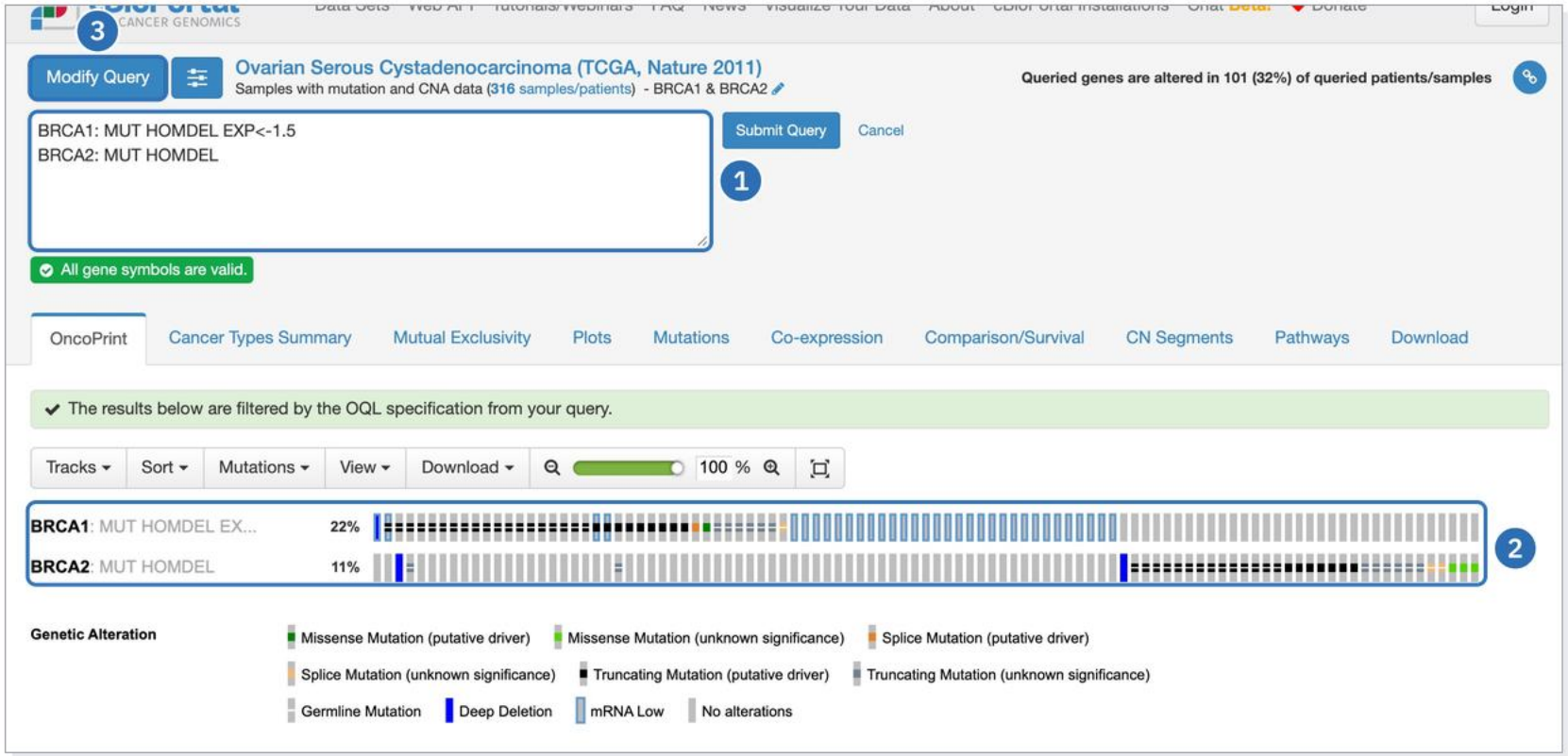
BRCA1/2: methylation and expression



1. Mutations and deletions are not the only way to lose a gene: DNA methylation can silence it. On the Plots tab, pick the “mRNA vs methyl” example.
2. Choose what to plot on each axis: here BRCA1 methylation against BRCA1 expression.
3. Samples with high methylation and low expression (lower right; Spearman -0.40). OQL can find them too: with low expression.

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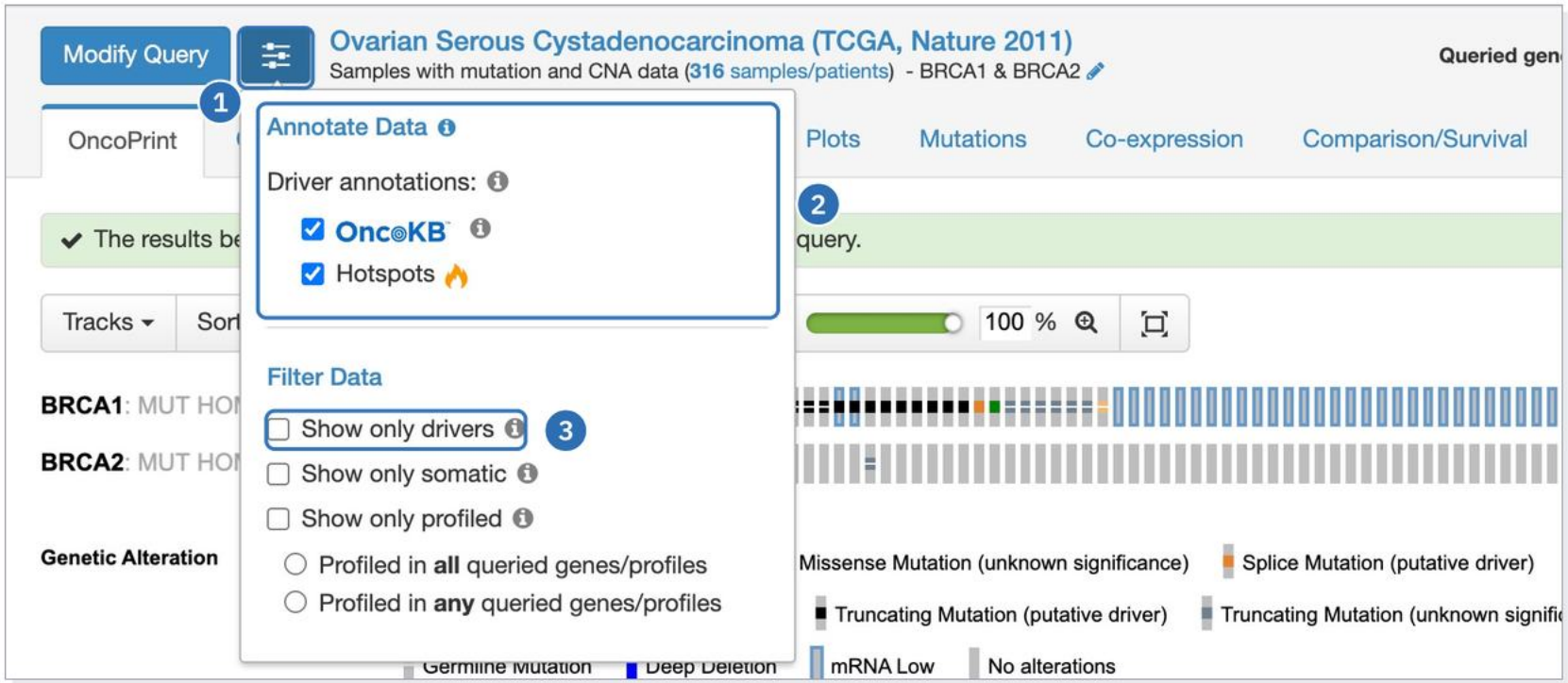
BRCA1/2: add low expression



1. BRCA1: MUT HOMDEL EXP<-1.5.
2. BRCA1 is now altered in 22% of samples (was 12%); low expression shows as a blue outline.
3. EXP needs an mRNA profile: use “Modify Query” to select mRNA expression in the query form. The gene edit box can't change the profiles.

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BRCA1/2: drivers and VUS

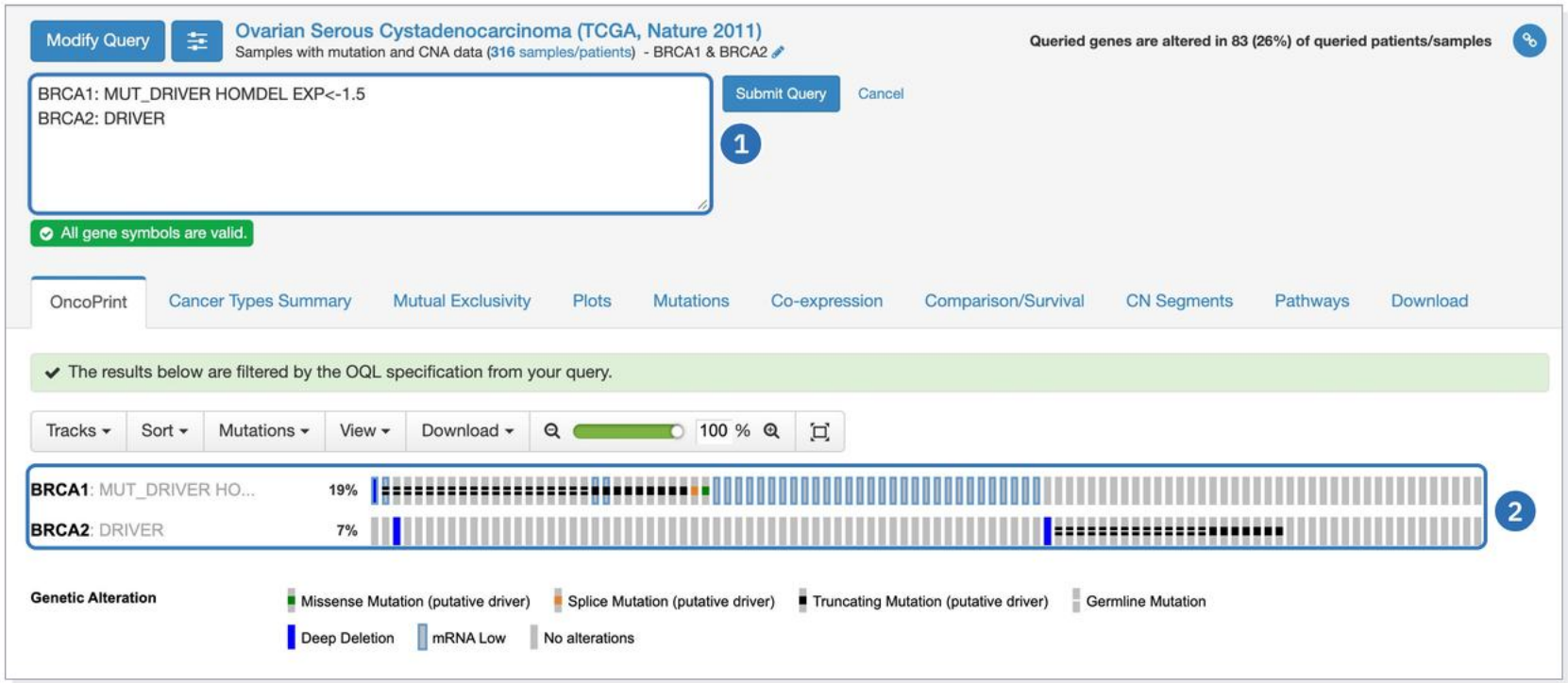


1. Some BRCA1/2 mutations are variants of unknown significance (VUS, light colors). The settings menu decides which mutations are putative drivers (see Tutorial 2).

2. The sources that decide which mutations count as putative drivers.

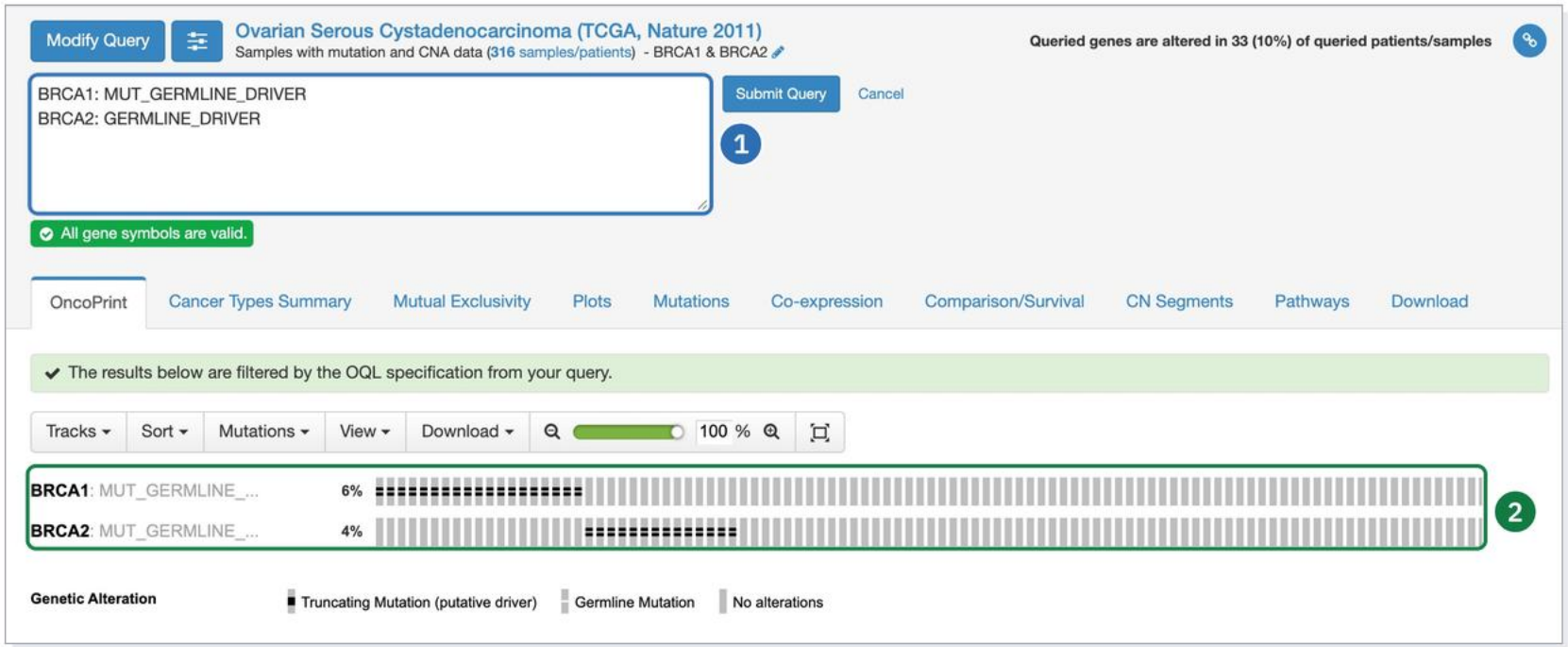
3. “Show only drivers” hides VUS on every tab. OQL can do it for one gene: the DRIVER modifier.

BRCA1/2: only drivers



1. BRCA1: MUT_DRIVER HOMDEL EXP<-1.5 keeps only driver mutations of BRCA1. BRCA2: DRIVER keeps only driver alterations of any kind (mutations, structural variants, copy number).
2. BRCA1 19%, BRCA2 7%: the VUS are gone.

BRCA1/2: germline drivers



1. This study includes germline mutations. MUT_GERMLINE_DRIVER (BRCA1) and GERMLINE_DRIVER (BRCA2) mean the same here: GERMLINE only applies to mutations.

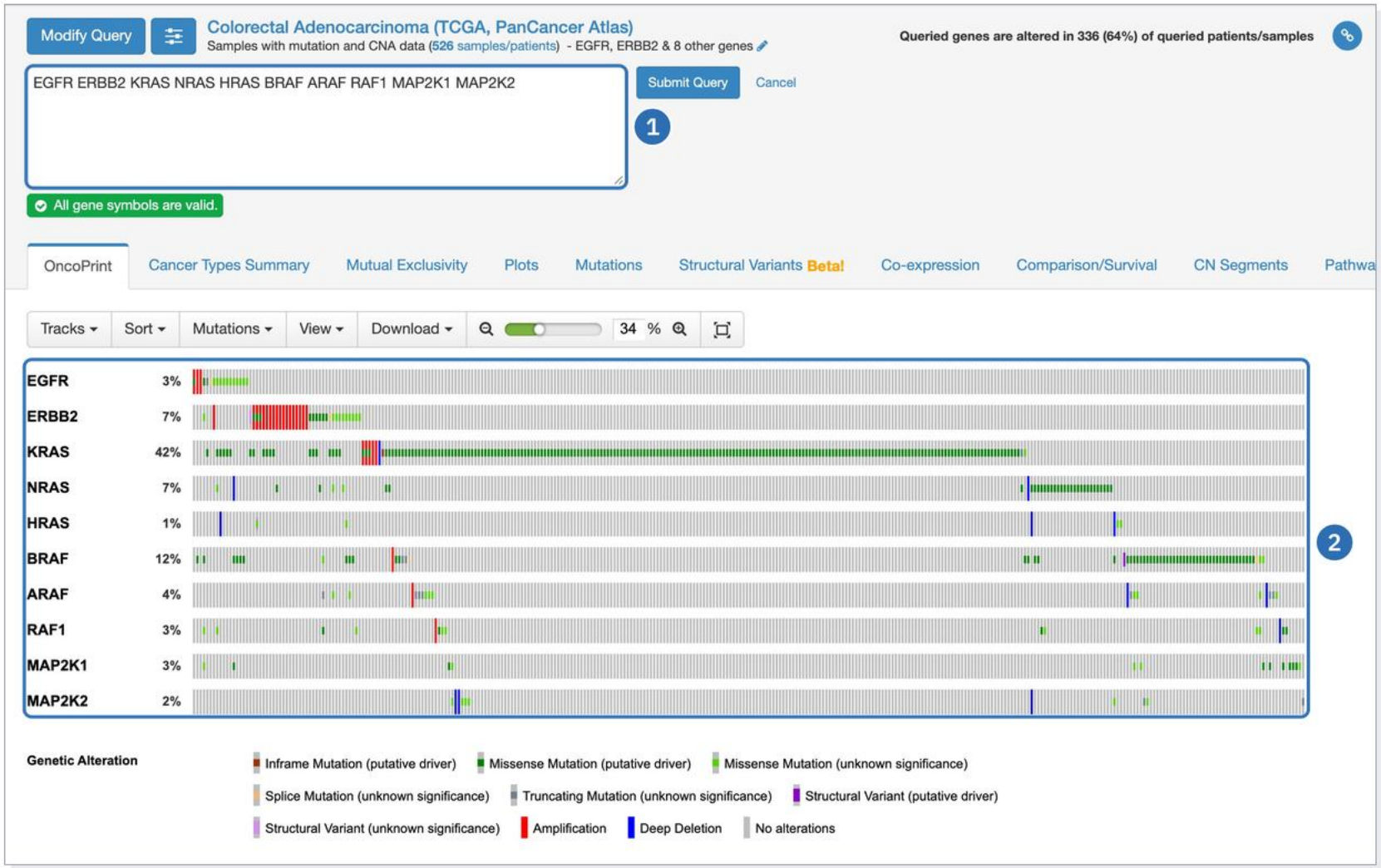
2. Q: How many samples have a germline driver mutation in BRCA1 or BRCA2?
   A: 10%: BRCA1 6%, BRCA2 4%.

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OQL example: RTK pathway alterations in colorectal cancer

- Receptor tyrosine kinases (EGFR, ERBB2) activate RAS (KRAS, NRAS, HRAS), which activates RAF (BRAF, ARAF, RAF1), which activates MEK (MAP2K1, MAP2K2).
- How are alterations spread across the levels of this pathway? Query all ten genes in Colorectal Adenocarcinoma (TCGA, PanCancer Atlas).

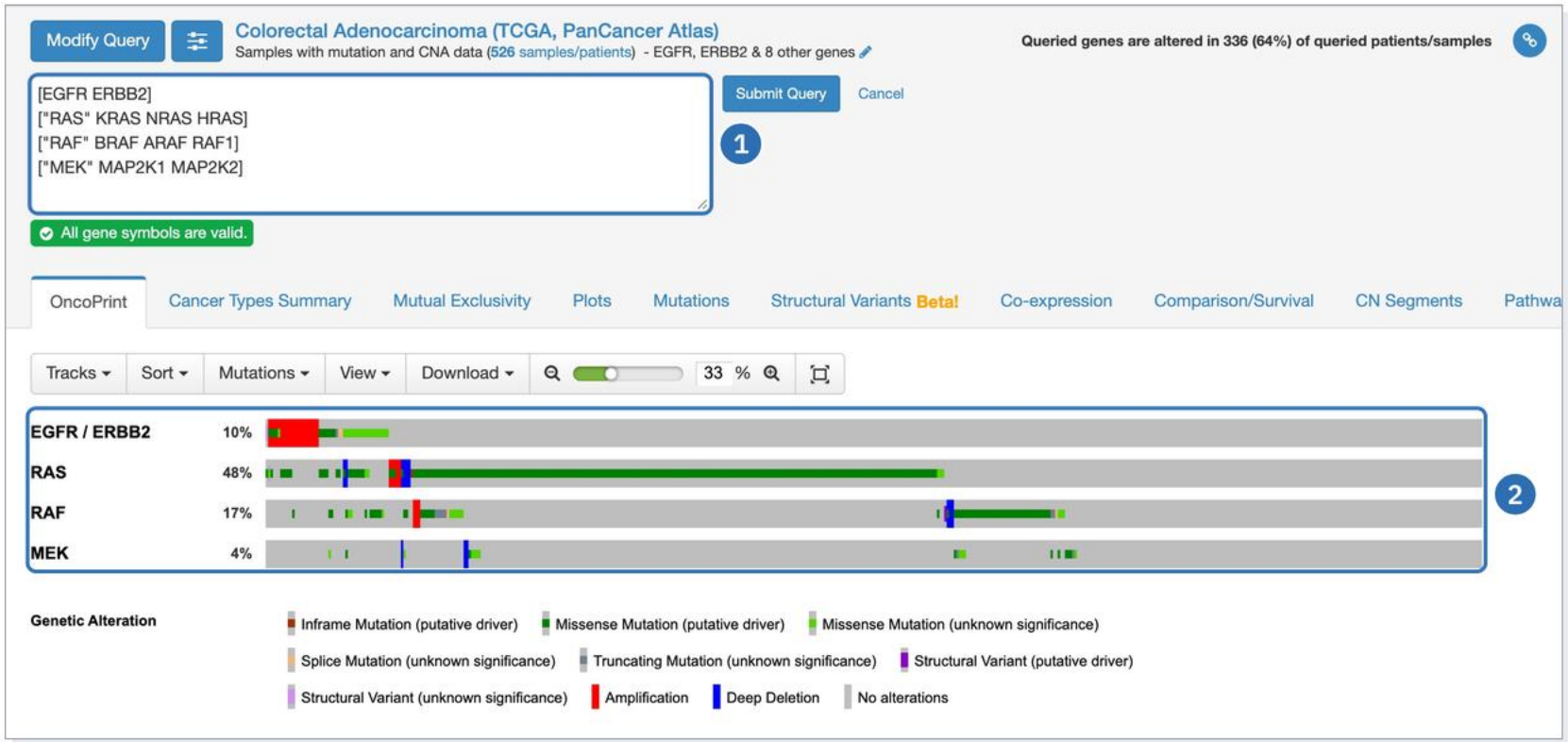
RTK pathway: one track per gene



1. All ten pathway genes, no OQL: one track per gene.
2. One row per gene: 64% of samples are altered but in this view it is hard to see the alteration frequency at the pathway level.

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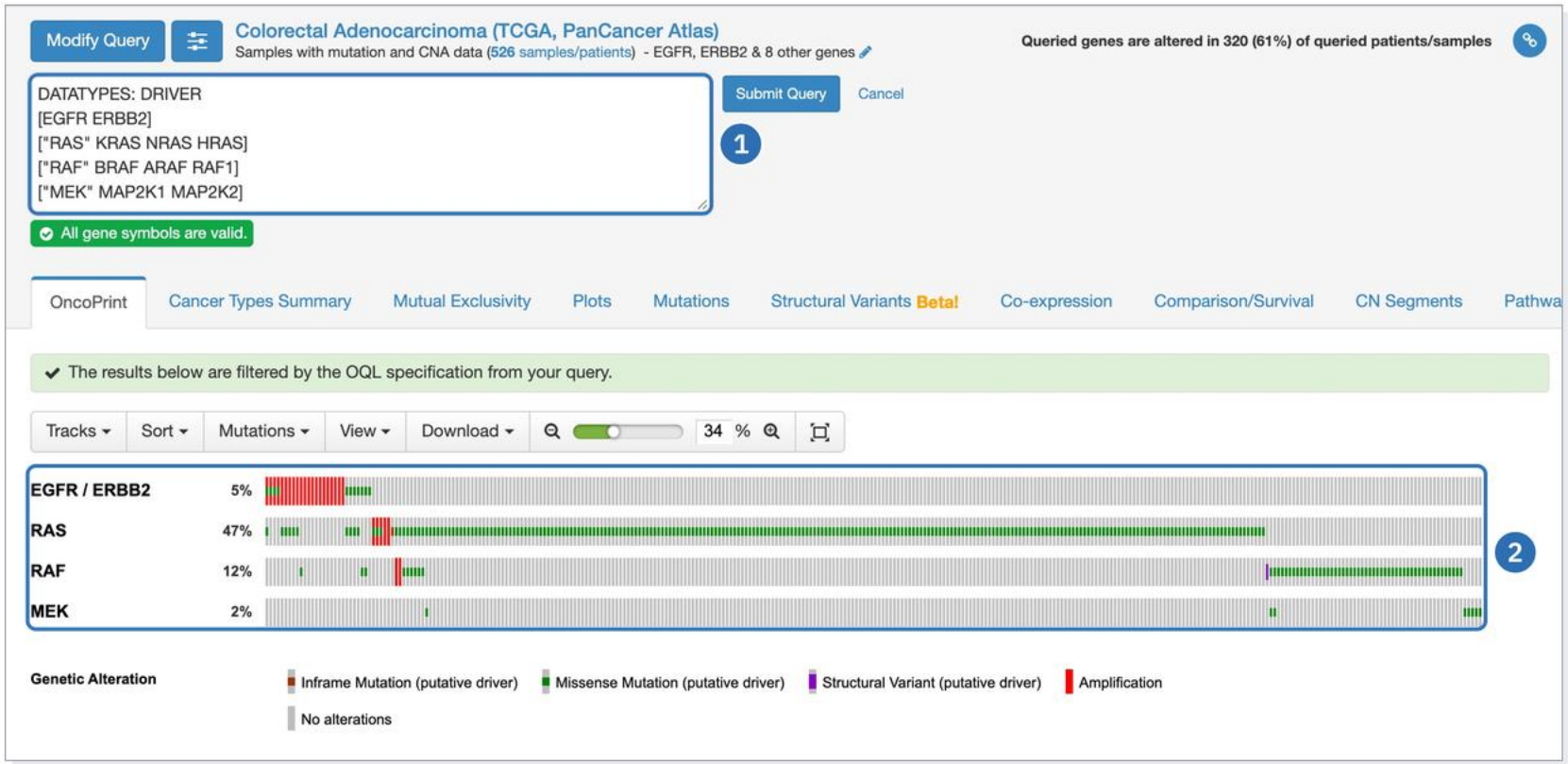
RTK pathway: gene tracks



1. Square brackets merge genes into one track: ["RAS" KRAS NRAS HRAS]. The name in quotes is optional (the first track is named EGFR / ERBB2).
2. One row per pathway level: RTK 10%, RAS 48%, RAF 17%, MEK 4%.

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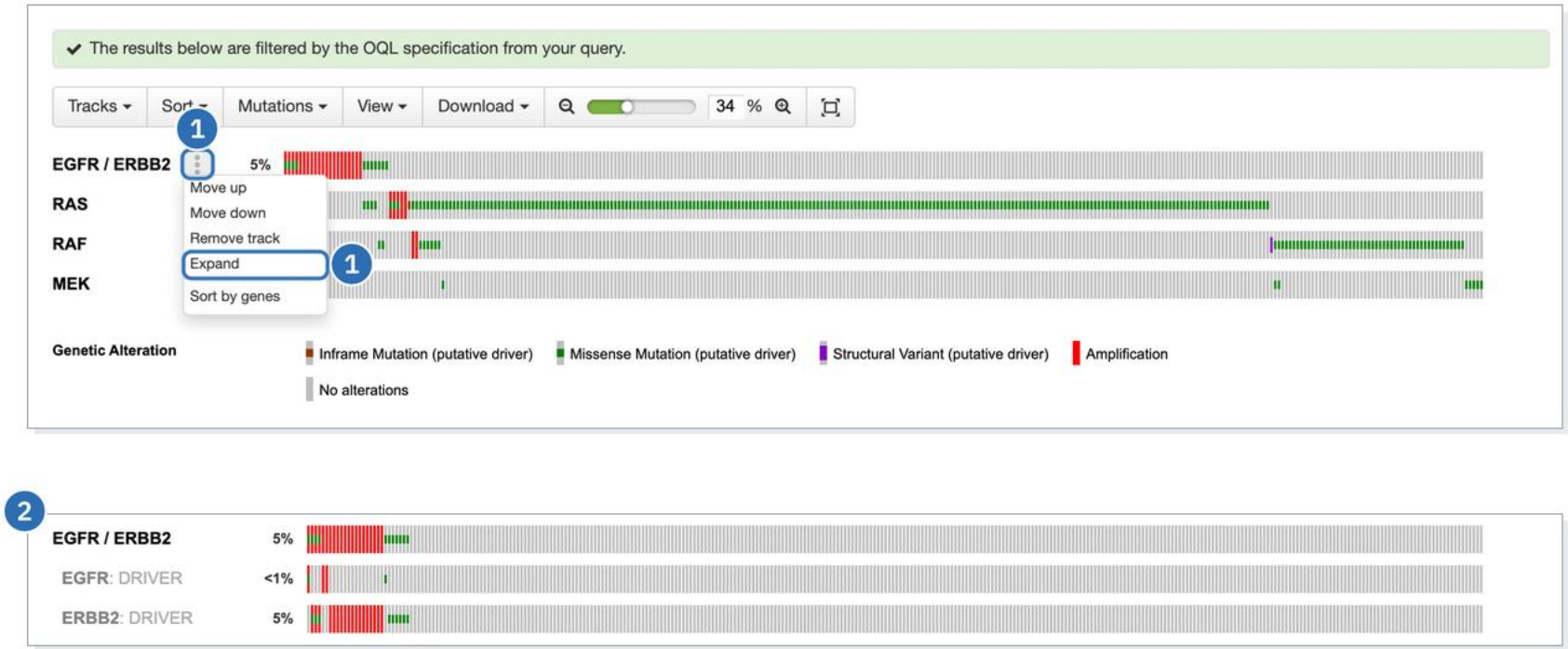
RTK pathway: gene tracks with OQL



1. DATATYPES: DRIVER applies to every gene in the tracks. OQL can also go inside the brackets.
2. Only driver alterations (61% of samples): they rarely hit two levels of the pathway in the same sample.

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RTK pathway: expanding a gene track



After Expand

1. Open a gene track's menu (the three dots) and choose “Expand”.
2. The genes in the track get their own rows (EGFR <1%, ERBB2 5%). Gene tracks work on the OncoPrint, Mutual Exclusivity and Comparison/Survival tabs; other tabs show the single genes.



Questions?

OQL specification: docs.cbioportal.org/user-guide/oql

More tutorials: cbioportal.org/tutorials

Email us: cbioportal@googlegroups.com